



BSc (Hons) in Information Technology
Year 3 - Semester 2 (2026)
IT3060 - Human Computer Interaction

Lab Sheet 4

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Year: 3

Semester: 2

Batch: Weekday

Group: 2.1

Activity 1: Define Usability Goals

1.1 Usability Objectives

Goal	Success Metric	Target	Business Alignment
New users can complete the sign-up without confusion and get their first AI plan	Task completion rate and time on task	$\geq 90\%$ completion and ≤ 3 min	Directly targets the 68% onboarding drop off reported in the case study
Users can quickly find and start today's AI workout from the home panel	Task completion rate and number of taps	100% success; ≤ 2 taps	Closes the personalization gap theme and supports retention
Users can log meals with minimal effort	Completion time and error count	≤ 20 sec and 0 critical errors	Addresses high friction nutrition tracking (P1, P5)
Users can read and understand their progress at a glance	Single Ease Question score	$\geq 5.5/7$	Addresses poor progress visualization/ waning motivation
Users can join a community challenge and understand the privacy setting	Task completion rate and Single Ease Question score	$\geq 85\%$ success and $\geq 5/7$	Addresses social isolation while respecting privacy concerns (P5)
Overall perceived usability of the redesigned prototype	System Usability Scale	≥ 80 (case study benchmark: 68 - > 87)	Supports the business goal of raising app store ratings from 3.8 back above 4.5

1.2 Alignment with Business Problems

- The above objectives were set in a measurable manner to ensure a direct connection between the results of usability testing and FitFlow's key business issues. These business issues include declining user retention rates, a drop in app store rating from 4.6 to 3.8 stars, and verification of new features based on AI, social, and nutrition features.

- Research and design sessions focused on three key issues previously identified:
 1. **Navigation efficiency** - whether users can easily navigate when planning and logging AI workouts.
 2. **Dashboard/ Progress usability** - users can easily see their progress and efforts, thereby reducing user churn.
 3. **Onboarding flow** - the initial experience when new users enter the app. This was identified as a major factor affecting 68% of dropouts.

1.3 Prioritization

Priority	Goal Area	Reason
High	Onboarding completion	Largest quantified pain point (68% drop off)
High	Starting today's AI workout	Core value proposition and affects daily retention
High	Nutrition logging effort	Most frequently cited frustration in interviews
Medium	Progress/dashboard clarity	Strong driver of motivation but not a hard blocker
Medium	Community/ privacy comprehension	Important but affects a smaller subset of sessions

Activity 2: Prepare Test Tasks and Success Criteria

2.1 Task Scenarios

ID	Scenario Given to Participant	Success Criteria	Time Limit	System Tested
T1	You've just installed FitFlow for the first time. Set up your profile and	Complete onboarding and reach the Home Dashboard with a	<= 3 min	Onboarding flow (prototype)

	get your first workout plan.	generated AI plan visible		
T2	You have 20 minutes before a meeting. Find and start today's recommended workout.	Reach the workout screen and tap Start within 2 taps from Home	≤ 20 sec	Home Dashboard: Workout Planner (prototype)
T3	You missed yesterday's session. Reschedule it to today.	Successfully reschedule the missed session using the plan screen	≤ 45 sec	Workout Planner (prototype)
T4	You just had lunch. Log your meal using the fastest method available.	Successfully log a meal via the camera button	≤ 20 sec	Nutrition Logger (prototype)
T5	Check how consistent you have been this month.	Locate and correctly interpret the streak/weekly chart	≤ 30 sec and SEQ ≥ 5	Progress Tracking (prototype)
T6	Join a community step-challenge with people you trust.	Join a challenge and correctly state who can see their activity	≤ 45 sec	Community Feed (prototype)
T7	You are worried about privacy. Turn off visibility of your workouts to anyone outside your circle.	Locate and correctly set the visibility/privacy toggle	≤ 40 sec	Community Feed: Settings (prototype)
T8	(Baseline task) Using the current, live FitFlow app, find your workout history from last week.	Locate historical workout log in the existing app	≤ 60 sec	Current FitFlow app (existing/live)

2.2 Test Script

Instructions for Participants

“Thank you for participating in this testing session. I will give you some short and realistic scenarios that can be performed using the FitFlow app. Try to say as much as possible about what comes to your mind while solving the task. What do you see? What do you expect? How do you feel about it? Don’t be afraid to make mistakes, because we are testing the app, not you. You can leave the session at any time. Do you have any questions before we start?”

Task testing procedure

Before starting each task, the moderator read the relevant scenario aloud to the participant. The purpose of the task was explained to the participant, but no additional instructions were given on how to solve the task. The aim was to allow the participant to use the app in a way that felt natural to them.

The timer was started when the participant made the first tap of the task and stopped when the task's success criterion was met or the given time limit expired.

In short: The participant was not told how to perform the task. He/ she was only given the scenario and observed how he/ she naturally used the app. The timer measured from the first tap to the success/time limit.

2.3 Post-Task Survey Questions

- **Single Ease Question (SEQ):** “Overall, how difficult or easy was this task for you?” (1 = very difficult, 7 = very easy) After completing each task, participants were asked to rate how easy or difficult the task was on a 1–7 scale. “If there was any confusion in this task, please describe it.” (Open ended)
- **Open-ended question:** “If there was any confusion in this task, please describe it.” If there was any confusion or ambiguity during the task, participants were given the opportunity to describe it.
- **Expectation question:** “What did you expect to happen when you tapped on [X]?” When the participant followed the wrong path, they were asked what they expected to happen when they tapped on [X].
- **System Usability Scale (SUS):** After completing the entire usability testing session, participants completed the 10-question SUS questionnaire.
- **Closing open-ended question:** “What is the one thing you would change first?” At the end of the testing session, participants were asked what they considered to be the first thing that needed to be changed.

Activity 3: Recruit Participants

3.1 Participant Mix

ID	Type	Age	Fitness Level	Device
P1	Current user	29	Intermediate	Android
P2	Current user	34	Beginner	IOS
P3	Current user	26	Advanced	IOS
P4	Lapsed user	24	Beginner	Android
P5	Lapsed user	31	Intermediate	Android
P6	Lapsed user	27	Beginner	IOS

P7	New/ non user	33	Intermediate	IOS
P8	New/ non user	22	Beginner	Android
P9	New/ non user	38	Beginner	Android
P10	Fitness trainer (secondary stakeholder)	30	Advanced	IOS

3.2 Screening Criteria

- It was mandatory to be 18 years of age or older and to be an owner/user of an Android or IOS smartphone suitable for using the FitFlow app.
- Participants were divided into 3 active FitFlow users, 3 inactive users (users who have stopped using FitFlow in the past 6 months), and 3 non-users of the FitFlow app.
- To avoid biasing the results towards any one user group, participants with different self-reported fitness levels were included. That is, participants were selected to represent beginner, intermediate, and advanced fitness levels.
- According to the FR7 (Trainer Adapted Design) requirement, at least one fitness trainer was required to be included among the secondary stakeholders.

3.3 Recruitment Channels

- A recruitment message was displayed to a group of users in the app to identify participants who were willing to participate in usability testing.
- Participants were recruited by sending email and SMS messages to existing users who had consented to receive research-related messages at the time of registration.
- A short 5-question survey was administered to universities and local gyms to recruit participants (non-users) who were not using the FitFlow app.
- To recruit fitness trainer participants, direct messages were sent to partner gyms to connect a fitness trainer for usability testing.

3.4 Screening Questionnaire (Summary)

- Before scheduling a session, a short online survey consisting of 5 questions was used to confirm the participant's device type, FitFlow user status, fitness level, and availability.

3.5 Scheduling and Incentives

- Each testing session was allocated 45 minutes. Of this, 30 minutes were used for usability testing and 15 minutes as a buffer time to obtain consent and answer participant questions.

- Participants were given a thank-you voucher as an appreciation for their time. Participants were informed about this voucher prior to the testing session, and the voucher was not contingent on completing all tasks.
- Testing sessions were conducted via video calls with screen sharing. The procedure used for the usability testing sessions in the case study was followed.

Activity 4: Conduct Think-Aloud Testing

4.1 Session Protocol

1. At the beginning of the session, the participant was given an introduction and consent was confirmed using the same consent form used in Lab Sheet 1. A short warm-up question was then asked regarding the participant's exercise routine.
2. The participant was given an explanation of the Think-Aloud technique. Then, a practice trial was conducted using an unrelated application to familiarize the participant with the process of verbalizing thoughts.
3. Tasks T1 to T7 were performed using the interactive low/medium-fidelity prototype. Task T8 was performed using the currently used real live application. The sequence of tasks T1–T7 was followed with minor adjustments as necessary.
4. After each task was completed, the participant was given a Single Ease Question (SEQ) and asked a relevant test question.
5. After completing all tasks, participants completed the System Usability Scale (SUS) questionnaire. A final open-ended question was also asked.
6. Finally, participants were thanked for their participation. They were also given a certificate of anonymity in accordance with the Data Management Plan mentioned in Lab Sheet 1.

4.2 Session Protocol

- What are you looking at right now?
- What did you expect would happen there?
- Talk me through what you're thinking.
- The moderator avoids hinting at the correct path, avoids explaining features unless a task is fully abandoned, and notes the exact point of hesitation or error for later severity rating.

4.3 Sample Session Notes

Participant	Task	Observed Behavior	Quote
P4 (Lapsed)	T1 Onboarding	Stopped at 8 seconds on the fitness level because I wasn't sure what to choose.	"I don't know if I'm a beginner or intermediate, is there a description?"
P2 (Current)	T2 Start workout	Found the start button very quickly from the card on the home page	"Oh nice, it's right there, I didn't even have to think."
P8 (New)	T4 Nutrition log	First tapped the picture of the meal, then found the camera button, and was able to use it	"I thought I'd have to type it in, the camera thing is way faster."
P6 (Lapsed)	T6 Join challenge	Checked the privacy settings first before connecting.	"Before I join anything I want to know who can see this."
P10 (Trainer)	T3 Reschedule	The session dragged on successfully, but she asked if she could do this to a client	"Can I do this on a client's plan too, or only my own?"

Activity 5: Measure Task Success and Completion Time

5.1 Task-Level Metrics (prototype and baseline tasks, n = 10)

Task	Success Rate	Avg. Completion Time	Avg. Error Count	Avg. SEQ
T1 Onboarding	80%	2m 45s	1.4	5.1 / 7
T2 Start workout	100%	12s	0.1	6.8 / 7
T3 Reschedule	90%	38s	0.3	5.9 / 7
T4 Log meal (camera)	90%	17s	0.2	6.4 / 7
T5 View progress	100%	22s	0.1	6.1 / 7

T6 Join challenge	80%	40s	0.6	5.0 / 7
T7 Privacy toggle	70%	52s	0.9	4.6 / 7
T8 Baseline (current app)	60%	58s	1.6	3.8 / 7

5.2 System Usability Scale Scores by Participant

Participant	SUS Score
P1	82.5
P2	87.5
P3	90.0
P4	65.0
P5	70.0
P6	72.5
P7	77.5
P8	80.0
P9	75.0
P10	85.0

The Mean System Usability Scale (SUS) score of all 10 participants was reported to be 78.5. The target set in Activity 1 was 80 or higher (≥ 80). The score of 78.5 is very close to the target and is significantly higher than the unofficial baseline score obtained by participants for the live application. The low success rate on the T8 task in the live application and the Single Ease Question (SEQ) score of 3.8/7 is related to the navigation difficulty identified in the original case study.

5.3 Logging Method

All quantitative measures, i.e. number of tasks completed successfully (number of successes), time spent (time spent), and number of errors (number of errors), were recorded in real-time by a note-taker in an observer spreadsheet while the moderator conducted the usability test. Since the data was time-stamped with the screen recordings, the metrics could be re-examined and

confirmed at any time later. A short online questionnaire was used to obtain Single Ease Question (SEQ) and System Usability Scale (SUS) responses. The SEQ response was provided after the participant completed each task, and the SUS questionnaire was completed at the end of the entire testing session.

Activity 6: Record Usability Issues and Severity Ratings

6.1 Logging Method

ID	Issue Description	Screen	Severity	Frequency	Related Task
U1	The fitness selection is ambiguous and makes people hesitant to enter.	Onboarding	Major	6/10	T1
U2	The privacy/ visibility trust is not self-evident unless a deliberate search is made.	Community Feed/ Settings	Critical	5/10	T7
U3	There is no message that you have successfully joined a challenge.	Community Feed	Minor	3/10	T6
U4	The drag and drop gesture to re-record a time is not intuitive for first-time users	Workout Planner	Major	4/10	T3
U5	Baseline: Currently requires three taps to access existing workout history	Current live app	Major	6/10	T8
U6	The camera button icon can be linked to a shortcut to the photo gallery	Nutrition Logger	Minor	2/10	T4

6.2 Severity Rating Scale Used

- **Critical:** Prevents task completion or poses a privacy/trustworthiness risk must be resolved before deployment.
- **Major:** Causes significant uncertainty, errors or omissions, but the task can be completed anyway.
- **Minor:** Causes some confusion or inefficiency, but does not prevent task completion.
- **Improvement:** No usability issues, but participant suggestion.

6.3 Notes on Frequency and Linkage

Based on initial research feedback from P5 and HMW2, the U2 (privacy toggle discoverability) issue has been classified as a Critical issue, although its incidence rate is medium. This is because the inability to easily find privacy controls can impact user trust and the user's ability to choose the social interaction they want. U1 and U4 are the two issues with the highest incidence rates. Both of these issues are related to onboarding and first plan adjustment flows. This is consistent with the findings of the case study, where it was identified that a user's first app visit is a high-risk time for abandonment.

Activity 7: Prepare Improvement Recommendations

7.1 Prioritized Recommendations

ID	Addresses	Recommendation	Priority	Implementation Effort
R1	U2	Add a persistent, visible privacy indicator (e.g. a lock icon with 'Circle only') on every screen where activity is shareable, not just in settings	Critical	Low
R2	U1	Add one short example line under each fitness-level option (e.g. 'Beginner new to structured workouts')	High	Low
R3	U4	Replace the drag gesture with a visible 'Reschedule' button as a secondary, discoverable action alongside drag to reorder	High	Medium
R4	U5	Surface a 'recent workouts' shortcut on the Home Dashboard of the current app as an interim fix while the redesign is built	Medium	Low
R5	U3	Add a brief confirmation toast ('You joined the 7 Day Streak challenge') after joining a challenge	Medium	Low
R6	U6	Restyle the camera button with a clearer food/ camera icon and a short 'Scan meal' label	Low	Low

7.2 Key Patterns Identified

- The main reason for all issues identified as Critical/Major was discoverability. Although the prototype had necessary features such as privacy controls, rescheduling, and workout history, users were unable to easily find them in the time required.
- First-session flows, i.e. check-in and first plan interactions, were identified as the areas with the lowest success rates and Single Ease Question (SEQ) scores. This confirms that the prioritization done in Activity 1 was correct, especially the high priority given to the 'check-in' feature.
- Every task in the prototype yielded significantly better results when compared to the comparable baseline task T8 in the live app. This confirms that the redesign direction outlined in Lab Sheet 3 is moving in the right direction to improve the usability of FitFlow.

7.3 Quick Wins vs. Deeper Fixes

Recommendations R1, R2, R5, and R6 require very minor changes, such as simple changes to copy (text), an icon, or a single confirmation state. Therefore, these changes are identified as quick wins that can be implemented and released immediately.

Recommendations R3 (rescheduling) and R4 (workout history shortcut) require minor interaction or navigation changes. Therefore, they are planned to be implemented in the next design iteration rather than released immediately.

7.4 Final Usability Test Report Summary

In total, the redesigned FitFlow prototype was evaluated for usability testing using 10 participants and 8 tasks. The System Usability Scale (SUS) score was 78.5, which was close to the target score of 80. Furthermore, a success rate of over 90% was achieved for key tasks such as AI workout and nutrition logging, both of which showed a clear improvement when compared to the same baseline tasks in the live application currently in use.

Two issues were identified as Critical/Major during the testing process: the undiscoverable privacy toggle and the unclear fitness-level onboarding options. Therefore, it is recommended that these two issues be resolved before the next testing round. In addition, four other usability issues were identified, for which concrete fixes were identified that could be implemented with relatively little effort.

Combining these findings with the previous design and requirements analysis conducted in Lab Sheets 1 through 3 has provided the redesign team with a solid foundation to make validated and evidence-based design decisions for the next iteration of the FitFlow interface.